MIL-M-28787/161A(EC)
12 October 1983
SUPERSEDING
MIL-M-28787/161(EC)
2 November 1976

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

TERMINATOR, NETWORK, LINEAR, KEY CODE AKE

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic network terminator module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.337 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be AKE.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/161-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

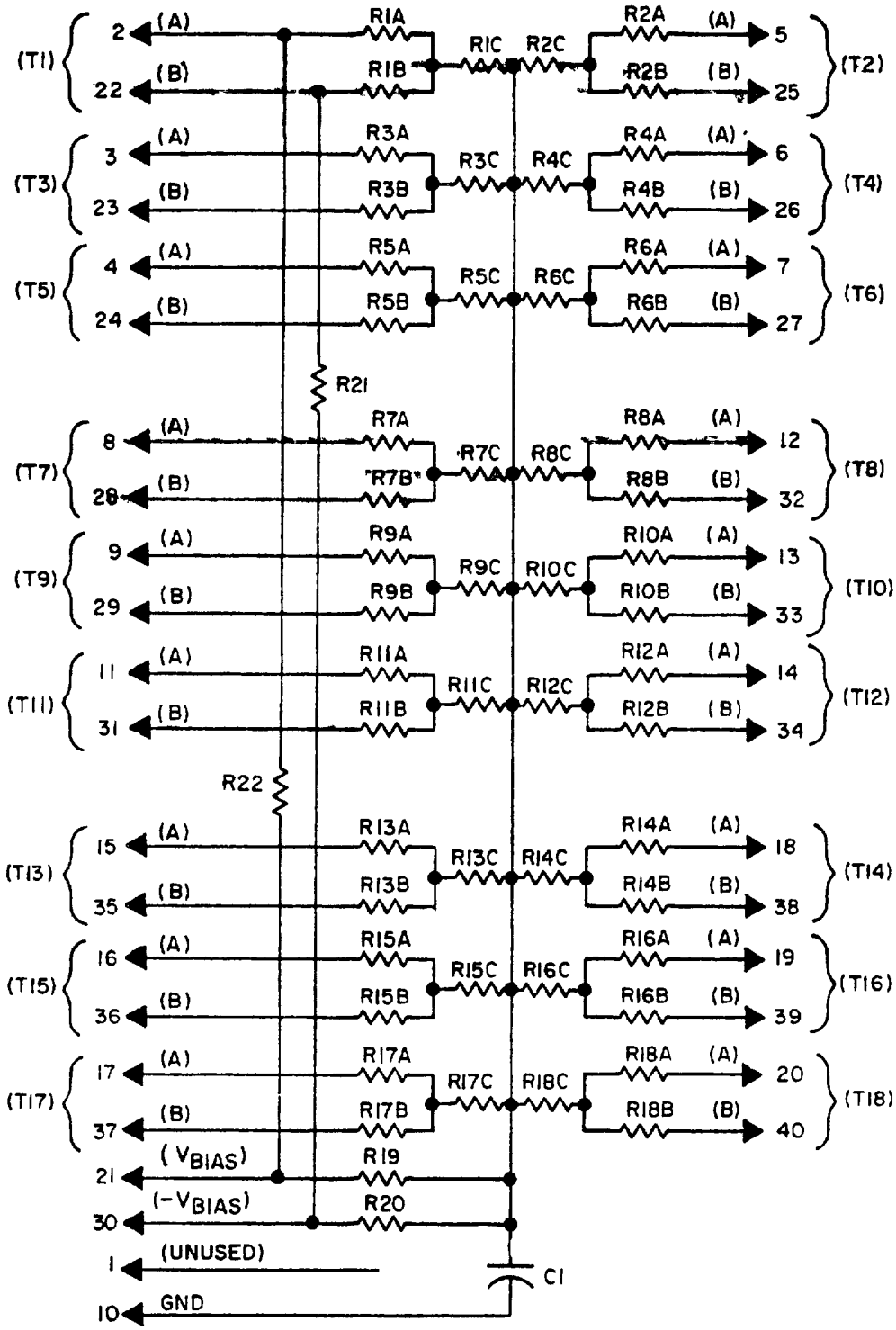
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. T1 = Biased terminator.
2. T2-T18 = Terminators.
3. V_{BIAS}, $-V_{BIAS}$ = Common bias inputs.
4. GND = 0 V Ref.
5. R19 = 220Ω, R20 = 91Ω, R21 = 150Ω, R22 = 470Ω.
6. C1 = 5.6 µF.

FIGURE 1. AKE module function diagram, network terminator.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1,A2	Isolation	II	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1	Terminator resistance	III	X	159	166	154	171	154	171	151	175	Ω
B2-B18	Terminator resistance	III	X	147	153	142	158	142	158	140	160	Ω
C1-C34	Common mode current	IV	X	15	21	14	22	15	21	14	22	mA
D	Differential offset voltage	V	X	1.07	1.50	1.02	1.50	1.02	1.50	0.98	1.50	V
E1-E34	Common mode bias voltage	VI	X	1.3	1.7	1.3	1.7	1.3	1.7	1.3	1.7	V
F1	Capacitor, 1 kHz	VII	X	240	410	240	410	210	430	210	430	mV rms
F2	Capacitor, 120 Hz	VII	X	1.5	2.5	1.5	2.5	1.3	2.7	1.3	2.7	V rms
G1-G18	Overshoot	VIII	Q	-300	300	-300	300	-300	300	-300	300	mV
H1-H18	Undershoot	VIII	Q	-300	300	-300	300	-300	300	-300	300	mV
J1-J18	Input crosstalk, high level	IX	Q	0	300	0	300	0	300	0	300	mV
K1-K18	Input crosstalk, low level	IX	Q	0	300	0	300	0	300	0	300	mV

1/ "X" parameters shall be 100 percent tested during acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through K shall be performed in any order with the following exceptions. If test condition A is part of a sequence of tests, the order of testing shall be as follows: A, where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through K in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through K shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B through K shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through K shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/161-1 supersedes part number M28787/161. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation	1,500 mW
Differential input voltage	5.5 V dc maximum
Differential impedance	150Ω
Common mode impedance T1	140Ω
Common mode impedance T2-T18	205Ω

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* 6.5 Module function. This module contains 18 terminators. Terminators T2-T18 have a differential impedance of 150Ω. Terminator T1 is a biased terminator with a differential impedance of 150Ω.

The 18 differential (and common mode) terminators are designed to terminate double-ended lines, driven by balanced drivers, which have a differential characteristic impedance Z_0 of 150Ω and a common mode impedance of $\approx$ 205Ω. The module contains an optional common bias network which can be used to bias all the signal lines to +1.5 V dc. To utilize the common bias network, V_{BIAS} (pin 21) is connected to a +5 V dc (nominal) supply and $-V_{BIAS}$ (pin 30) is connected to ground.

Terminator T1 contains a bias network which provides a differential offset voltage across the terminator with pin 2 being more positive than pin 22. The function of this biased terminator is to supply a differential input to a fully balanced receiver without a cable or drive input. This terminator can then be utilized to disable control lines logically which would be indeterminant when a fully balanced receiver is used and an input cable is not connected. The biased terminator is only operative when the common bias network described above is used. The bias network reduces the impedance of terminator T1 and thus requires more drive current if this terminator is used for logic signals.

6.5.1 Power dissipation. Each termiantor shall be capable of dissipating 200 milliwatts maximum. Total of terminators power dissipation shall not exceed the maximum module power dissipation. Power dissipation shall be limited to one watt maximum within each group of terminators specified below.

The terminators groups are:

a. T1-T6

b. T7-T12

c. T13-T18

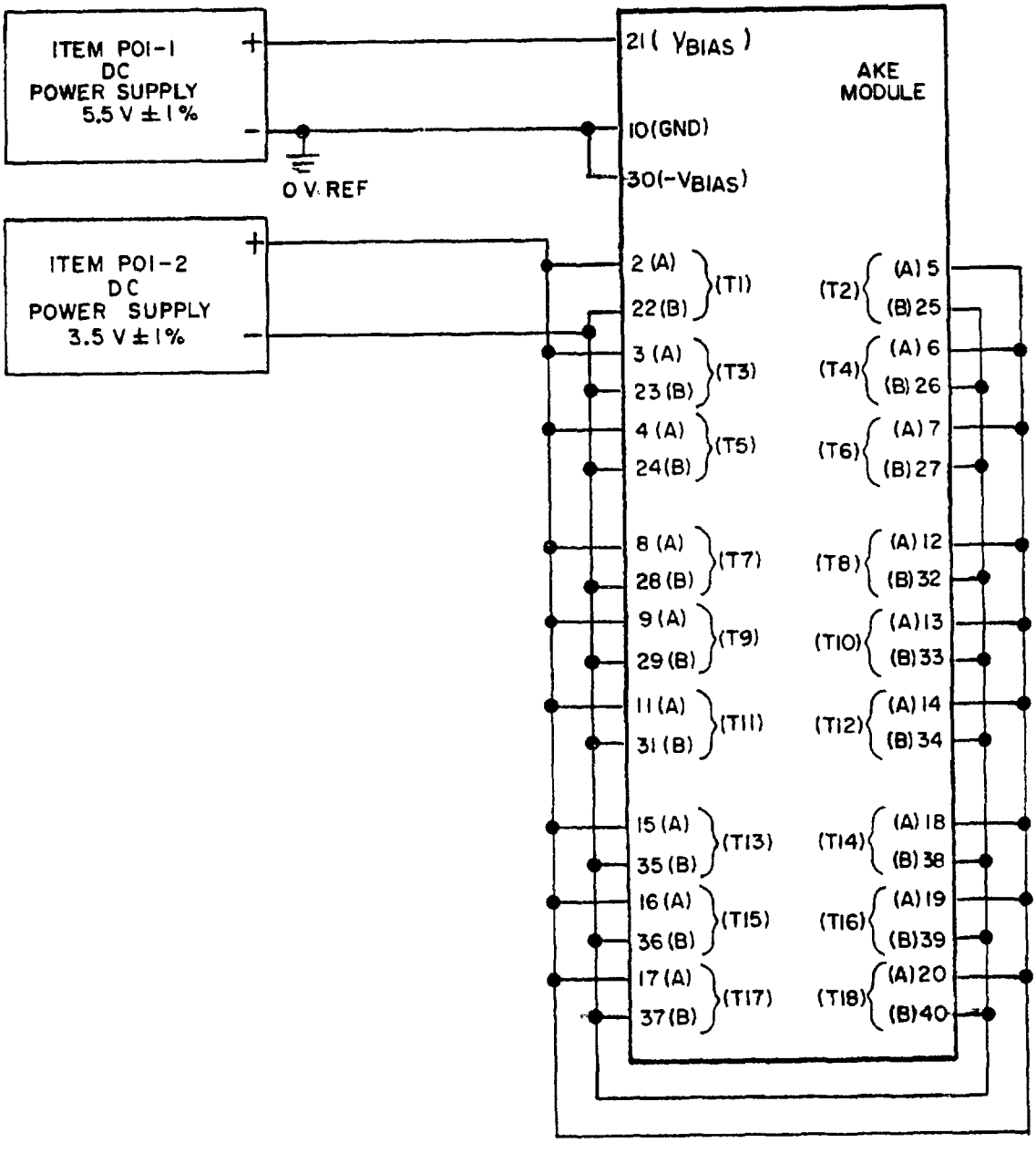
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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* FIGURE 2. Operating temperature and life test circuit.

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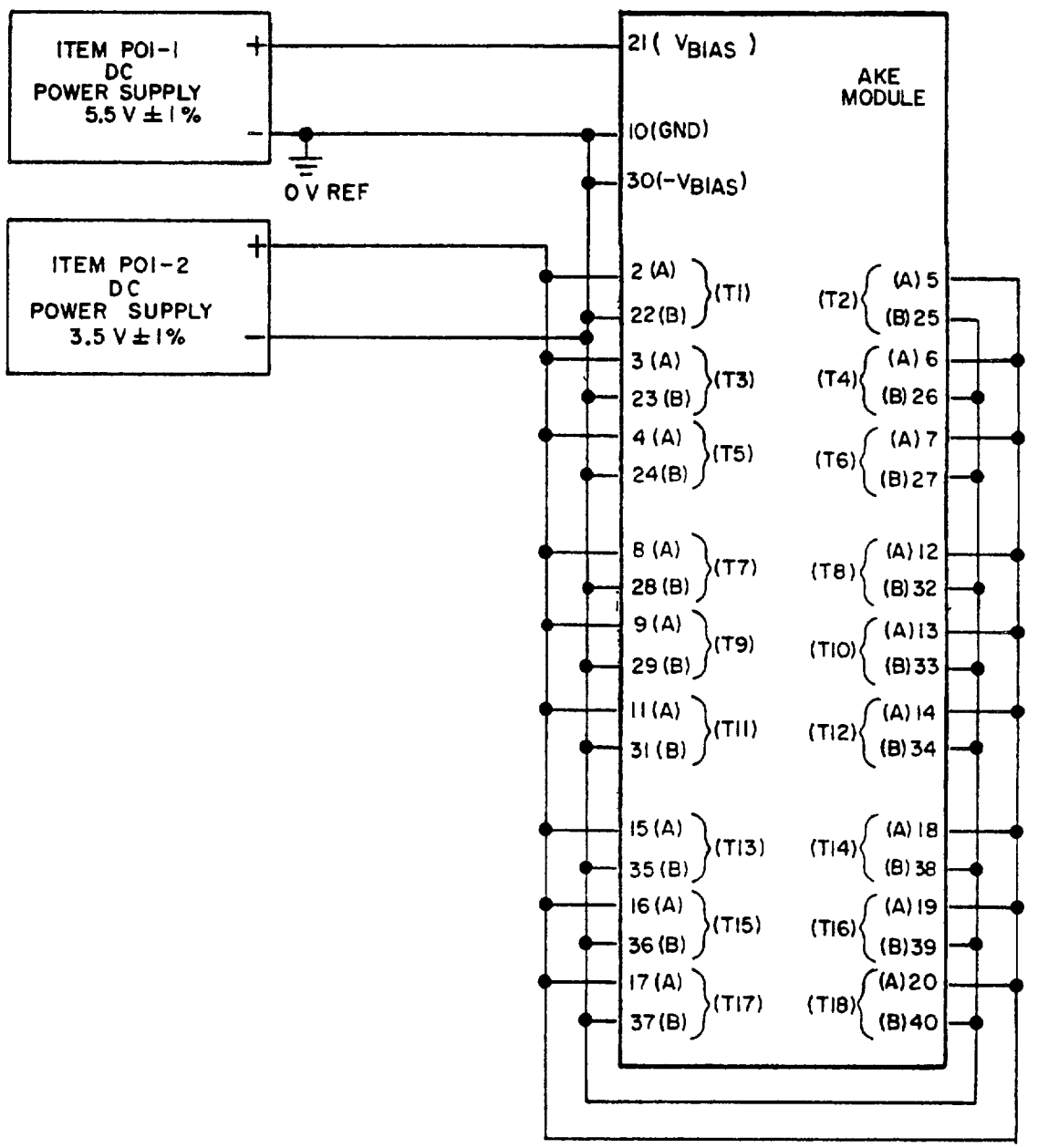


FIGURE 3. Component temperature test circuit.

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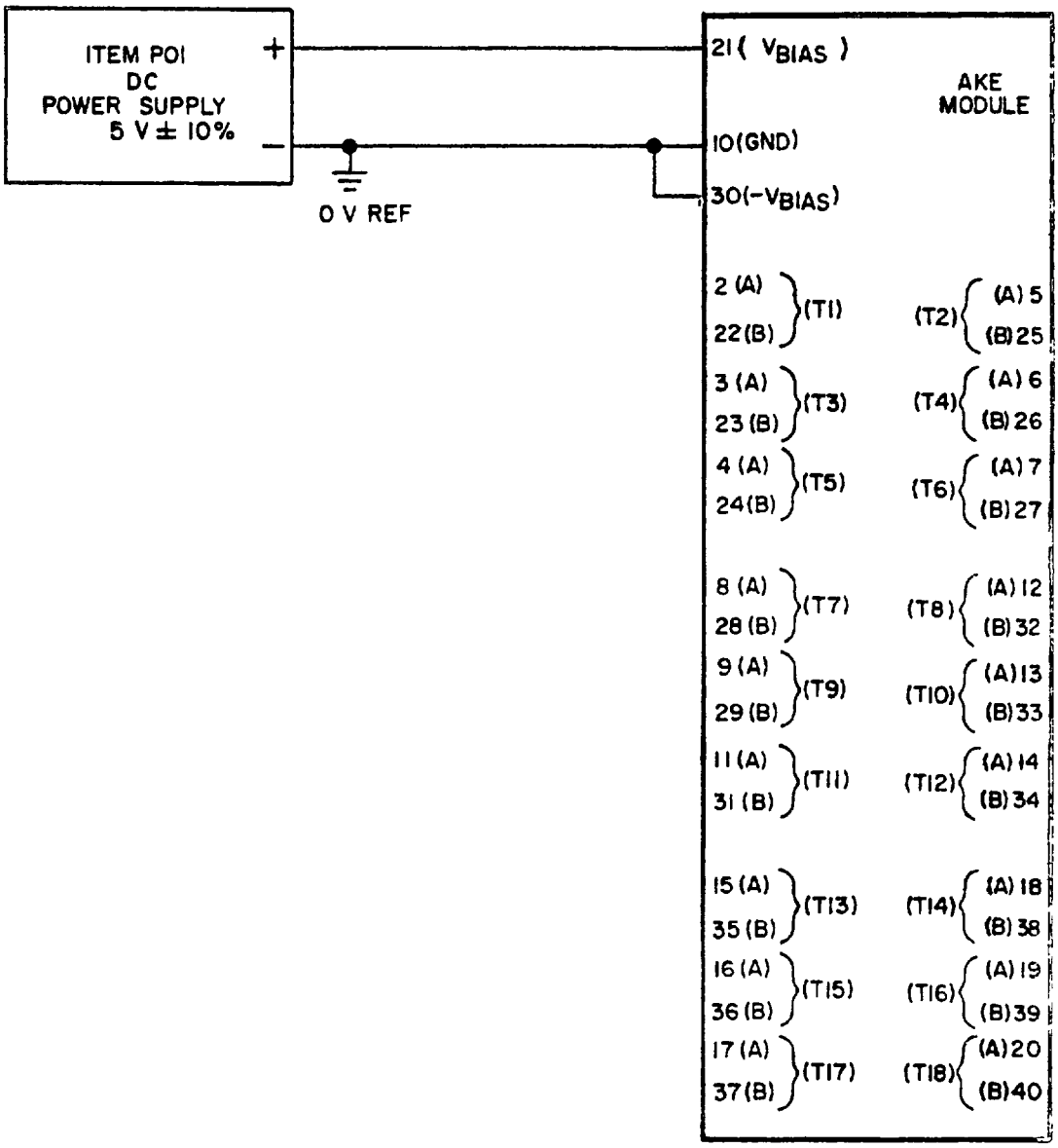


FIGURE 4. Vibration test circuit.

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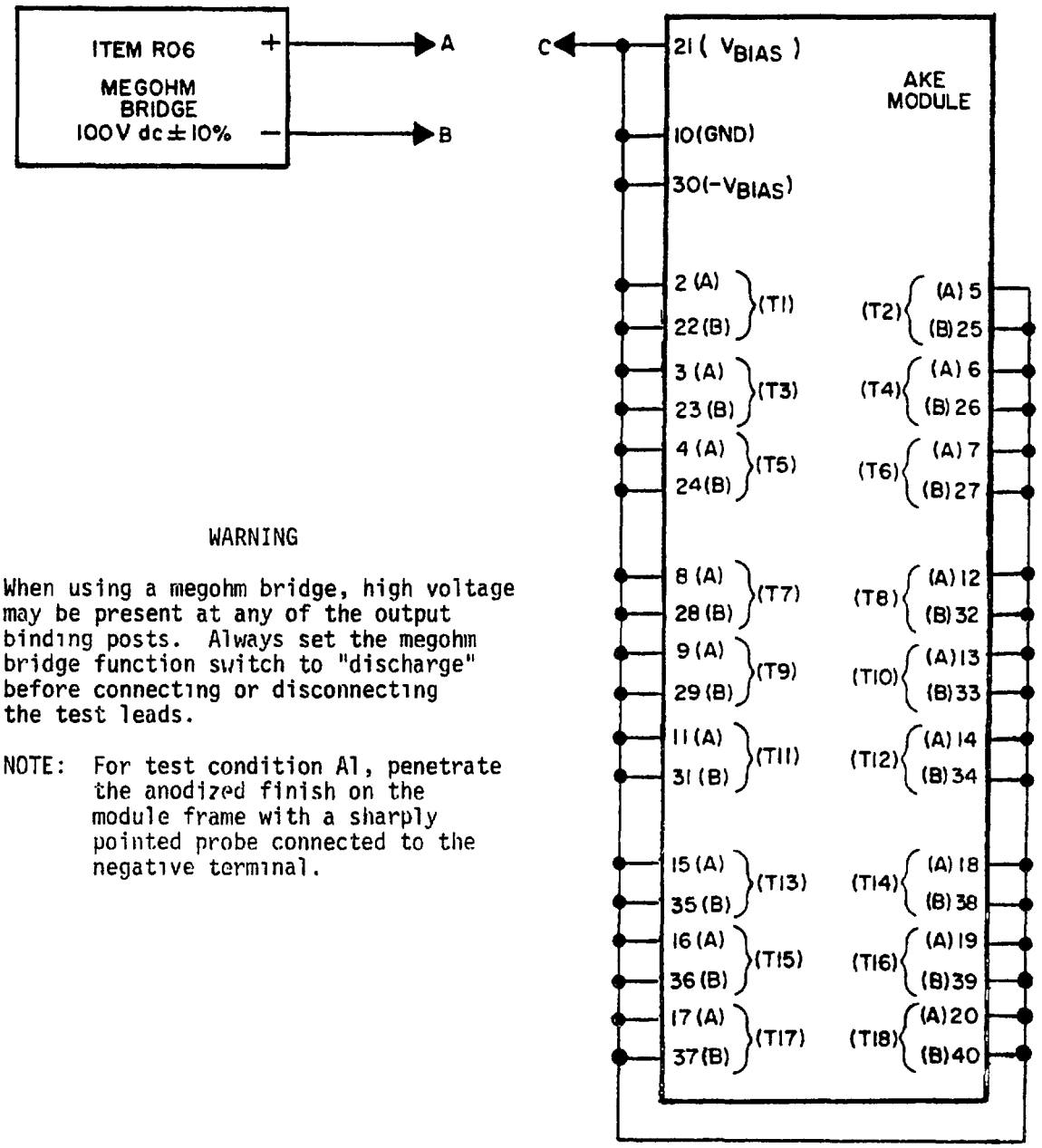
TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, frame	C, 1	Frame	10	∞	MΩ	5
A2	Isolation, used pins to unused pin	C	1	10	∞	MΩ	5

TABLE III. Terminator resistance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	T1 Terminator resistance	2	22	159	166	Ω	6
B2	T2 "	5	25	147	153	"	"
B3	T3 "	3	23	"	"	"	"
B4	T4 "	6	26	"	"	"	"
B5	T5 "	4	24	"	"	"	"
B6	T6 "	7	27	"	"	"	"
B7	T7 "	8	28	"	"	"	"
B8	T8 "	12	32	"	"	"	"
B9	T9 "	9	29	"	"	"	"
B10	T10 "	13	33	"	"	"	"
B11	T11 "	11	31	"	"	"	"
B12	T12 "	14	34	"	"	"	"
B13	T13 "	15	35	"	"	"	"
B14	T14 "	18	38	"	"	"	"
B15	T15 "	16	36	"	"	"	"
B16	T16 "	19	39	"	"	"	"
B17	T17 "	17	37	"	"	"	"
B18	T18 "	20	40	"	"	"	"

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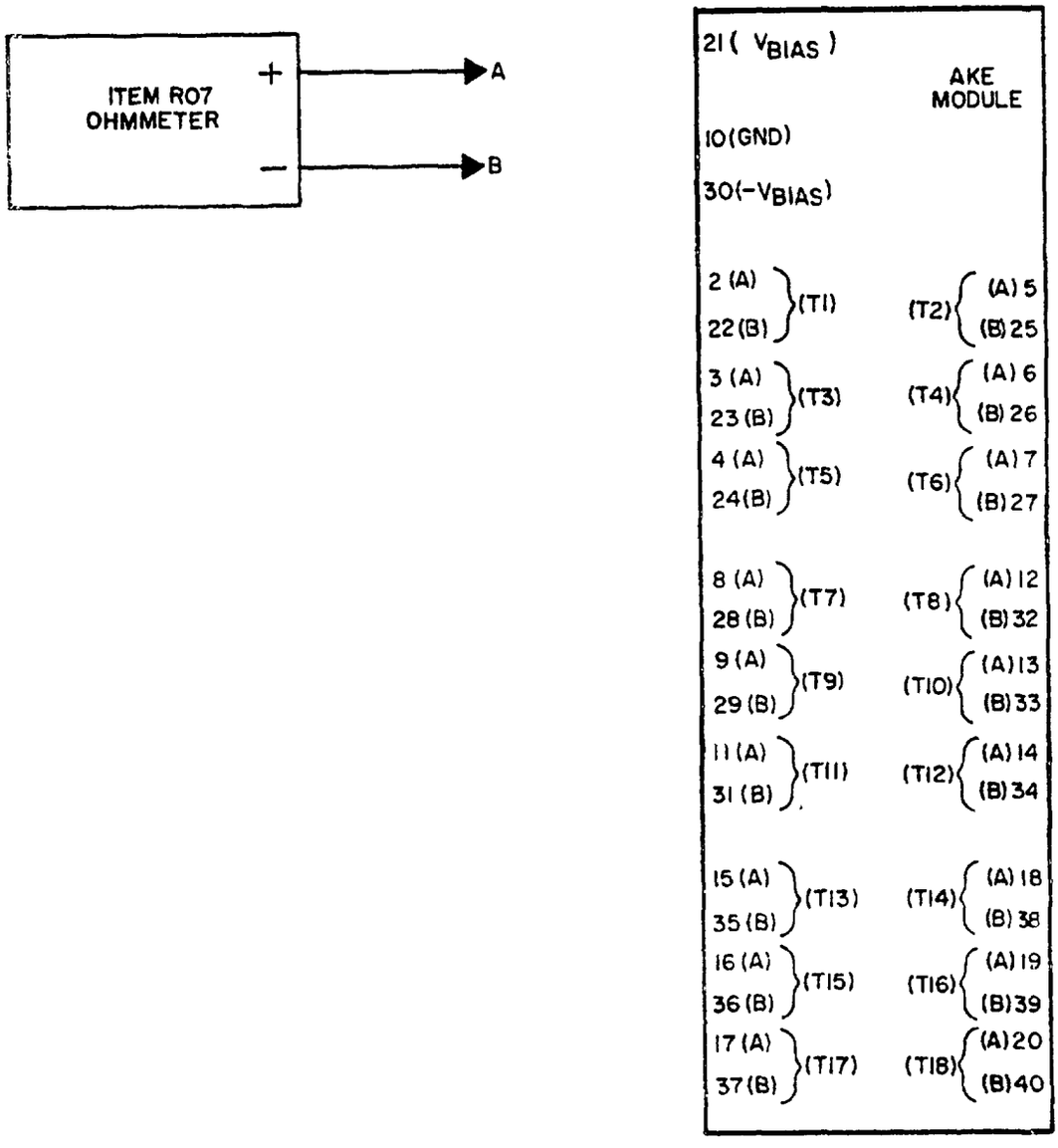
WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table II.

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* FIGURE 6. Terminator resistance test circuit for table III.

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TABLE IV. Common mode current test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
C1	T2A Common mode current	5	15	21	mA	7
C2	T2B "	25	"	"	"	"
C3	T3A "	3	"	"	"	"
C4	T3B "	23	"	"	"	"
C5	T4A "	6	"	"	"	"
C6	T4B "	26	"	"	"	"
C7	T5A "	4	"	"	"	"
C8	T5B "	24	"	"	"	"
C9	T6A "	7	"	"	"	"
C10	T6B "	27	"	"	"	"
C11	T7A "	8	"	"	"	"
C12	T7B "	28	"	"	"	"
C13	T8A "	12	"	"	"	"
C14	T8B "	32	"	"	"	"
C15	T9A "	9	"	"	"	"
C16	T9B "	29	"	"	"	"
C17	T10A "	13	"	"	"	"
C18	T10B "	33	"	"	"	"

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TABLE IV. Common mode current test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
C19	T11A Common mode current	11	15	21	mA	7
C20	T11B "	31	"	"	"	"
C21	T12A "	14	"	"	"	"
C22	T12B "	34	"	"	"	"
C23	T13A "	15	"	"	"	"
C24	T13B "	35	"	"	"	"
C25	T14A "	18	"	"	"	"
C26	T14B "	38	"	"	"	"
C27	T15A "	16	"	"	"	"
C28	T15B "	36	"	"	"	"
C29	T16A "	19	"	"	"	"
C30	T16B "	39	"	"	"	"
C31	T17A "	17	"	"	"	"
C32	T17B "	37	"	"	"	"
C33	T18A "	20	"	"	"	"
C34	T18B "	40	"	"	"	"

TABLE V. Differential offset voltage test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D	Differential offset voltage	2	22	1.07	1.50	V	8

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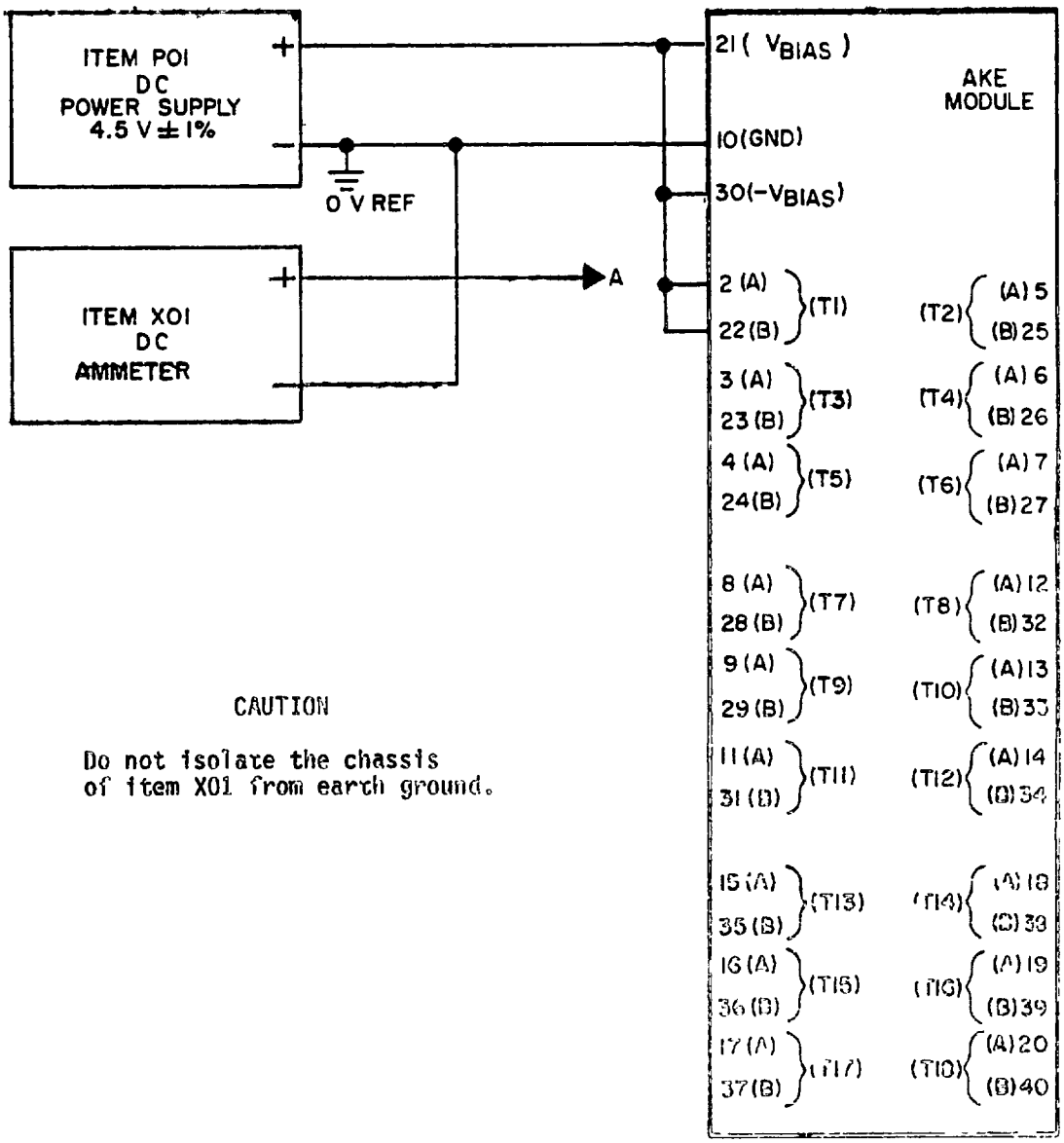
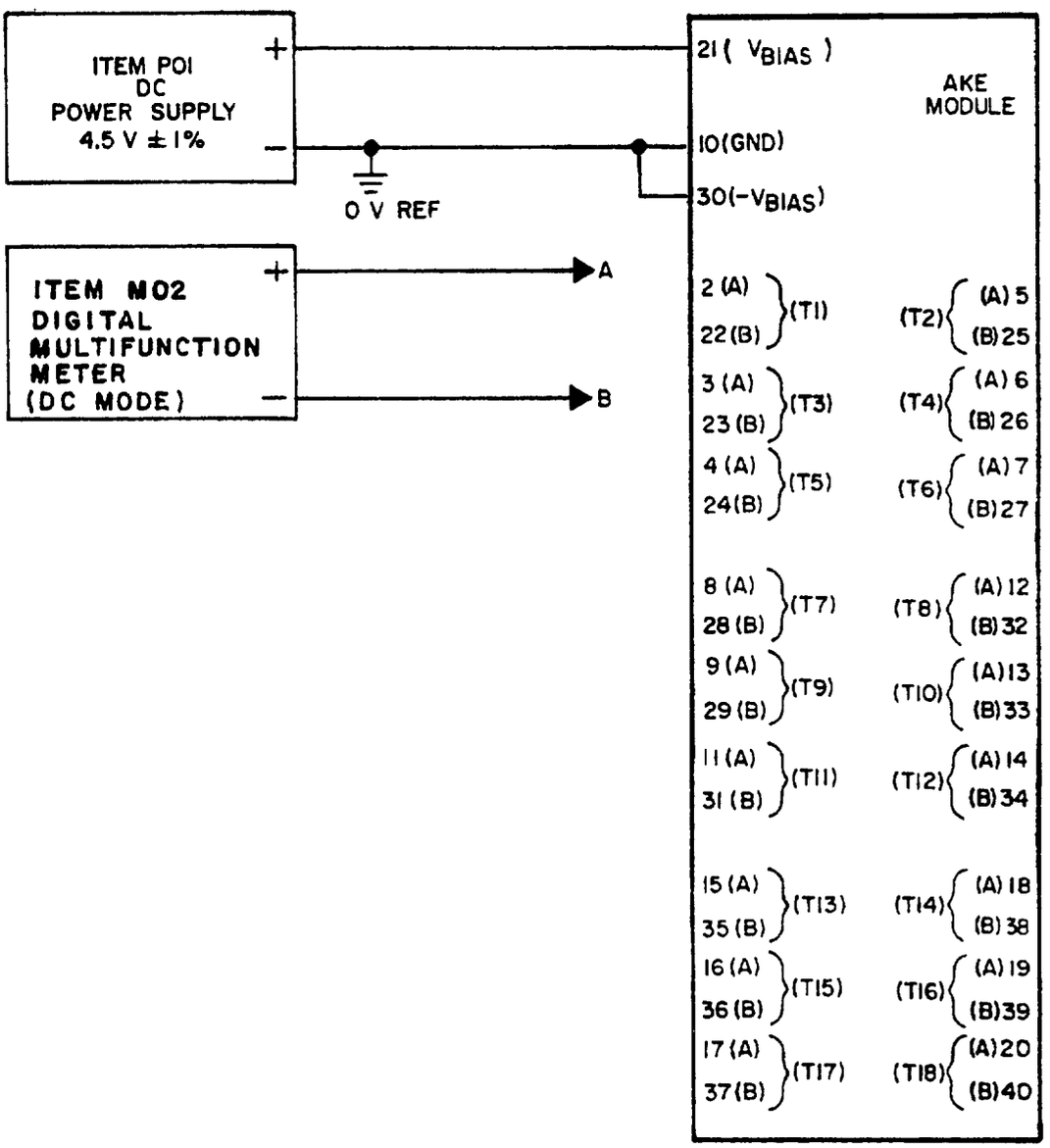


FIGURE 7. Common mode current test circuit for table IV.

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* FIGURE 8. Differential offset voltage test circuit for table V.

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TABLE VI. Common mode bias voltage test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E1	T2A Common mode bias voltage	5	1.3	1.7	V	9
E2	T2B "	25	"	"	"	"
E3	T3A "	3	"	"	"	"
E4	T3B "	23	"	"	"	"
E5	T4A "	6	"	"	"	"
E6	T4B "	26	"	"	"	"
E7	T5A "	4	"	"	"	"
E8	T5B "	24	"	"	"	"
E9	T6A "	7	"	"	"	"
E10	T6B "	27	"	"	"	"
E11	T7A "	8	"	"	"	"
E12	T7B "	28	"	"	"	"
E13	T8A "	12	"	"	"	"
E14	T8B "	32	"	"	"	"
E15	T9A "	9	"	"	"	"
E16	T9B "	29	"	"	"	"
E17	T10A "	13	"	"	"	"
E18	T10B "	33	"	"	"	"

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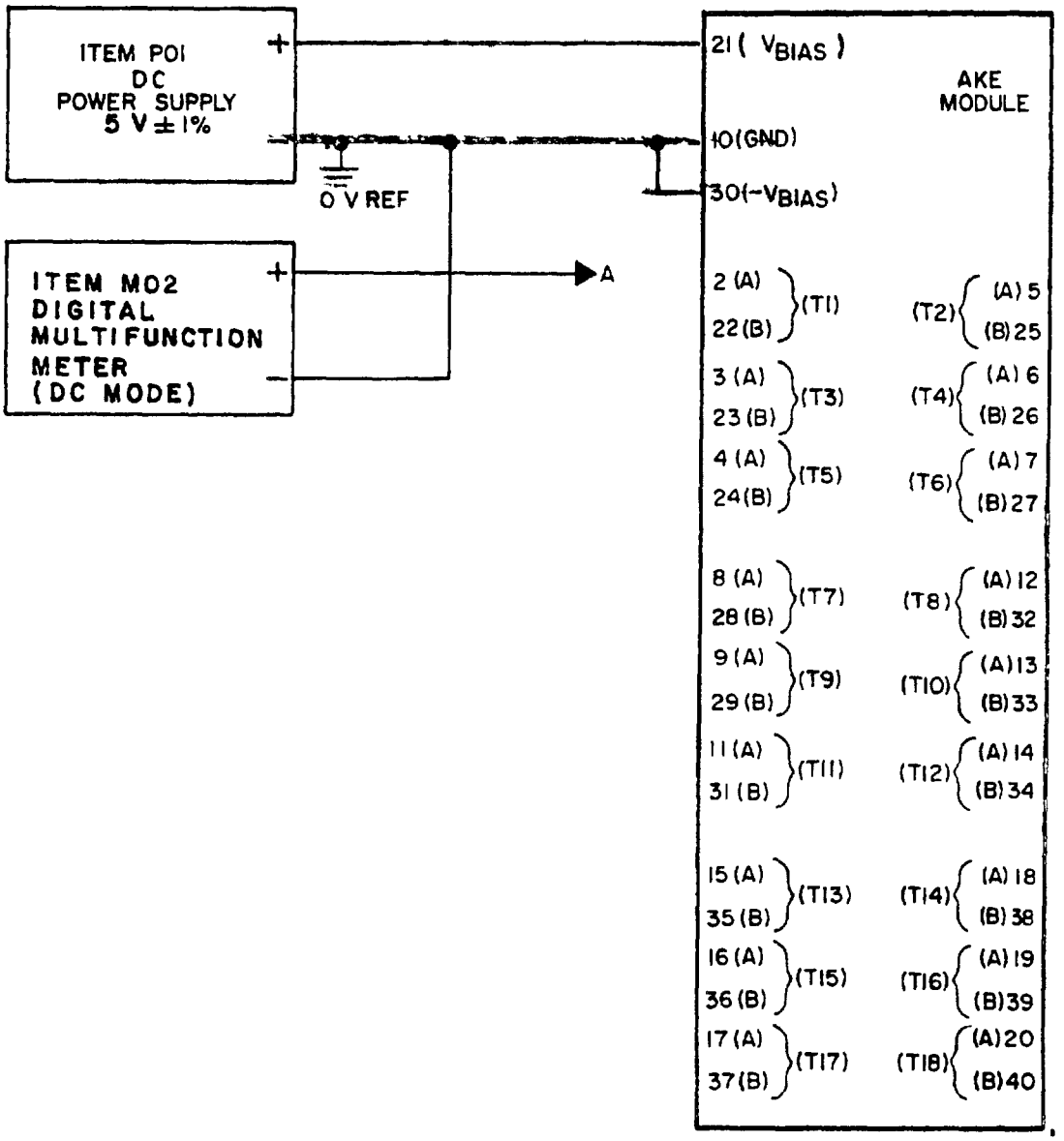
TABLE VI. Common mode bias voltage test - Continued.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
E19	T11A Common mode bias voltage	11	1.3	1.7	V	9
E20	T11B "	31	"	"	"	"
E21	T12A "	14	"	"	"	"
E22	T12B "	34	"	"	"	"
E23	T13A "	15	"	"	"	"
E24	T13B "	35	"	"	"	"
E25	T14A "	18	"	"	"	"
E26	T14B "	38	"	"	"	"
E27	T15A "	16	"	"	"	"
E28	T15B "	36	"	"	"	"
E29	T16A "	19	"	"	"	"
E30	T16B "	39	"	"	"	"
E31	T17A "	17	"	"	"	"
E32	T17B "	37	"	"	"	"
E33	T18A "	20	"	"	"	"
E34	T18B "	40	"	"	"	"

TABLE VII. Capacitor test.

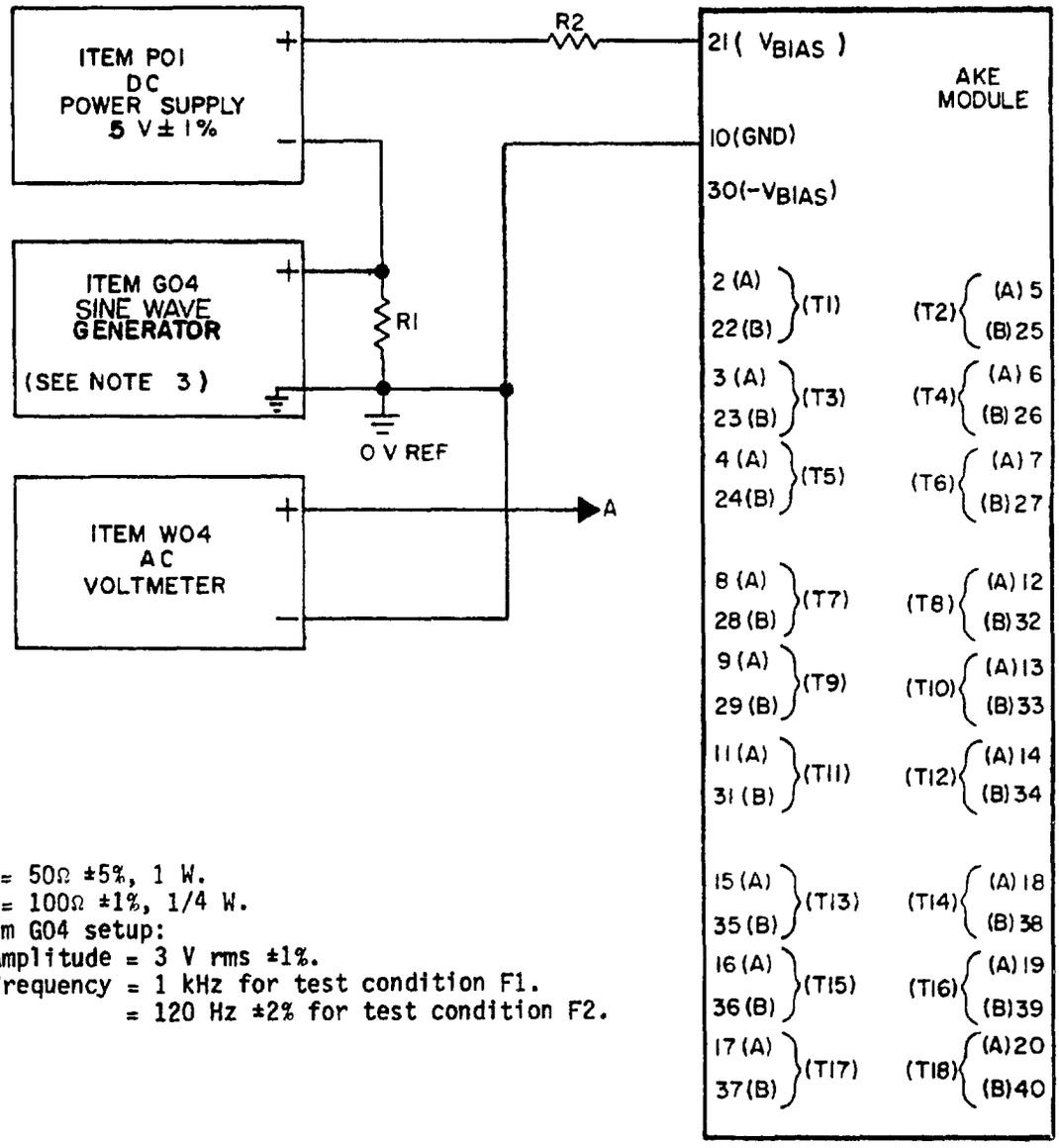
Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
F1	Capacitor, 1 kHz	5	240	410	mV rms	10
F2	Capacitor, 120 Hz	5	1.5	2.5	V rms	10

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* FIGURE 9. Common mode bias voltage test circuit for table VI.

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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = 100Ω ±1%, 1/4 W.
3. Item G04 setup:
   Amplitude = 3 V rms ±1%.
   Frequency = 1 kHz for test condition F1.
   = 120 Hz ±2% for test condition F2.

* FIGURE 10. Capacitor test circuit for table VII.

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TABLE VIII. Overshoot and undershoot test.

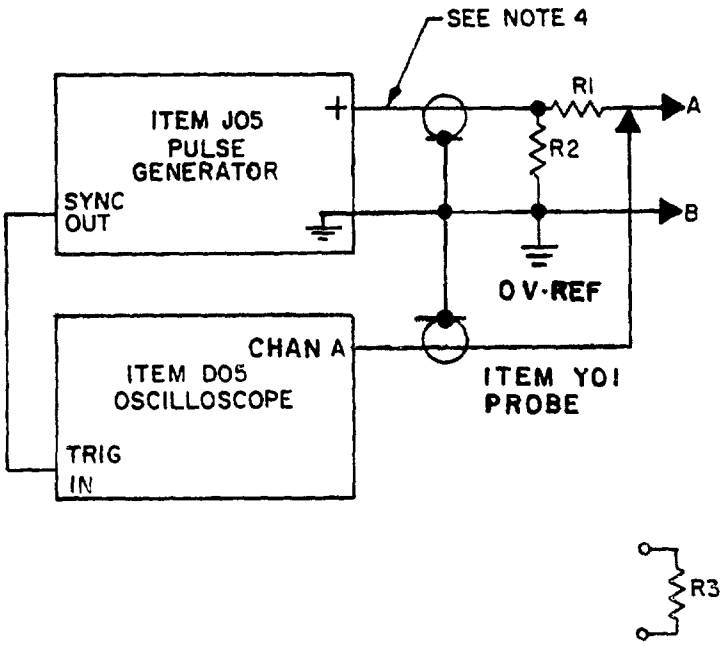
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
G1	T1 Overshoot	2	22	-300	300	mV	11,12
G2	T3 "	3	23	"	"	"	"
G3	T5 "	4	24	"	"	"	"
G4	T2 "	5	25	"	"	"	"
G5	T4 "	6	26	"	"	"	"
G6	T6 "	7	27	"	"	"	"
G7	T7 "	8	28	"	"	"	"
G8	T9 "	9	29	"	"	"	"
G9	T11 "	11	31	"	"	"	"
G10	T8 "	12	32	"	"	"	"
G11	T10 "	13	33	"	"	"	"
G12	T12 "	14	34	"	"	"	"
G13	T13 "	15	35	"	"	"	"
G14	T15 "	16	36	"	"	"	"
G15	T17 "	17	37	"	"	"	"
G16	T14 "	18	38	"	"	"	"
G17	T16 "	19	39	"	"	"	"
G18	T18 "	20	40	"	"	"	"

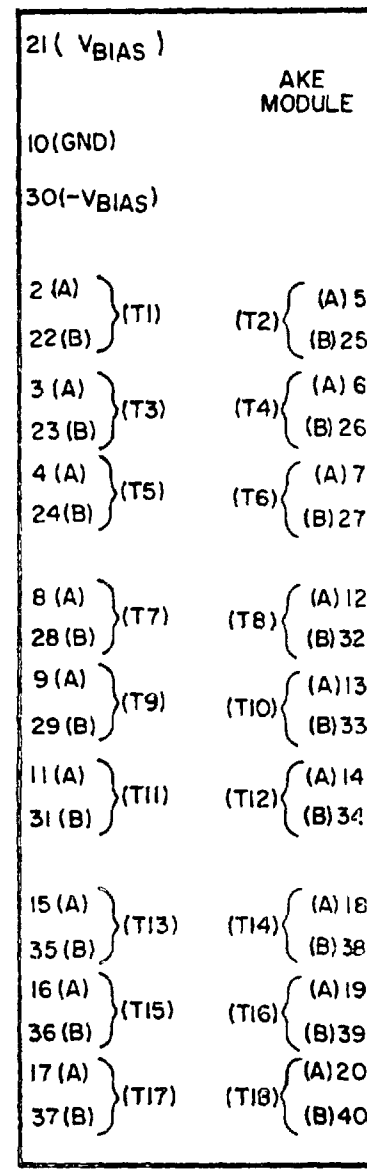
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TABLE VIII. Overshoot and undershoot test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	T1 Undershoot	2	22	-300	300	mV	11,12
H2	T3 "	3	23	"	"	"	"
H3	T5 "	4	24	"	"	"	"
H4	T2 "	5	25	"	"	"	"
H5	T4 "	6	26	"	"	"	"
H6	T6 "	7	27	"	"	"	"
H7	T7 "	8	28	"	"	"	"
H8	T9 "	9	29	"	"	"	"
H9	T11 "	11	31	"	"	"	"
H10	T8 "	12	32	"	"	"	"
H11	T10 "	13	33	"	"	"	"
H12	T12 "	14	34	"	"	"	"
H13	T13 "	15	35	"	"	"	"
H14	T15 "	16	36	"	"	"	"
H15	T17 "	17	37	"	"	"	"
H16	T14 "	18	38	"	"	"	"
H17	T16 "	19	39	"	"	"	"
H18	T18 "	20	40	"	"	"	"

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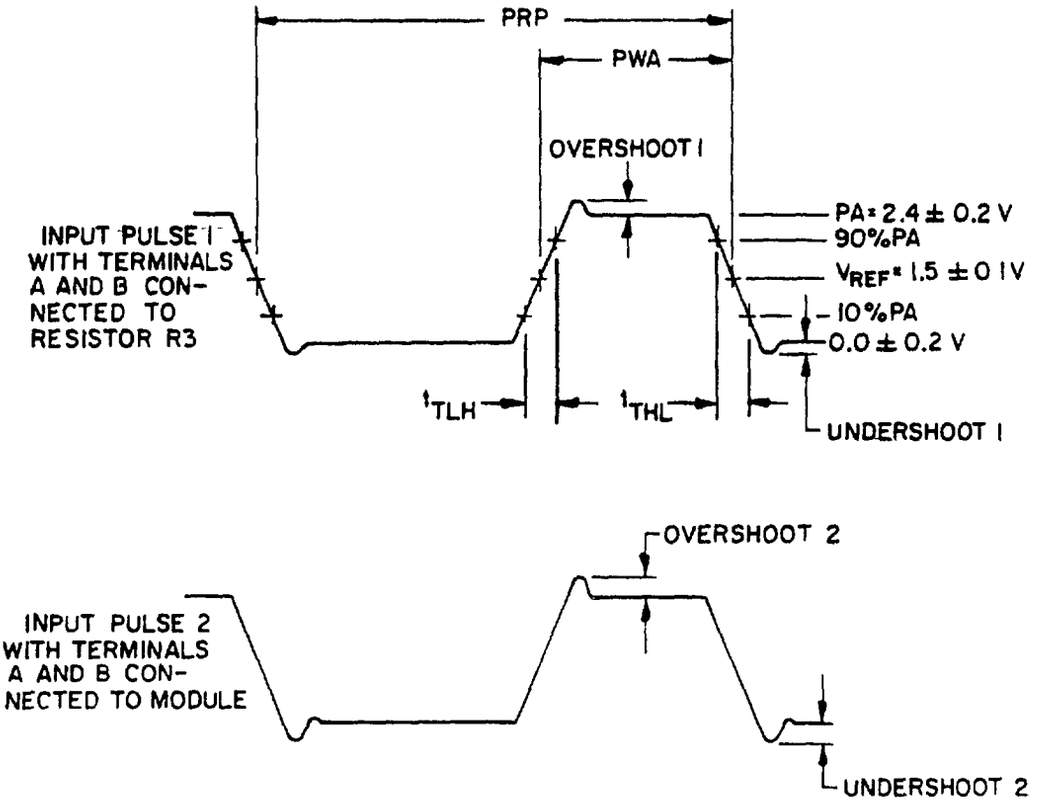




NOTES:
1. R1 = 100Ω ±1%, 1/4 W.
2. R2 = 61.9Ω ±1%, 1/4 W.
3. R3 = 150Ω ±1%, 1/4 W.
4. Cable impedance shall match item J05 impedance (RG 316/U or equivalent).

FIGURE 11. Overshoot and undershoot test circuit for table VIII.

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NOTES:
1. Connect terminals A and B across resistor R3 and adjust item J05 for input pulse 1 parameters:

   (a) PRP = 1,000 ±200 ns.
   (b) PWA = 400 ±50 ns.
   (c) $t_{TLH} = t_{THL}$ = 4 ±1 ns.
   (d) Overshoot 1 = Undershoot 1 = 0.25 V maximum.
   (e) Measure and record overshoot 1 and undershoot 1.

2. With item J05 adjusted for input pulse 1 parameters per note 1, connect terminals A and B to module pins per table VIII. Measure and record overshoot 2 and undershoot 2 as shown for input pulse 2.
3. Calculate overshoot and undershoot.

   Overshoot = overshoot 2 - overshoot 1.
   Undershoot = undershoot 2 - undershoot 1.

4. Lead lengths from item J05 cable to points A and B; item Y01 to module pin; and from item Y01 ground to module pin shall be less than 1 inch (25.4 mm).

* FIGURE 12. Overshoot and undershoot test circuit waveforms for table VIII.

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TABLE IX. Input crosstalk (high-level and low-level) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
J1	T1 Input cross-talk, high level	22	2	23,24, 25,26, 27	3,4, 5,6, 7	A	0	300	mV	13,14
J2	T3 "	23	3	22,24, 25,26, 27	2,4, 5,6, 7	"	"	"	"	"
J3	T5 "	24	4	22,23, 25,26, 27	2,3, 5,6, 7	"	"	"	"	"
J4	T2 "	25	5	22,23, 24,26, 27	2,3, 4,6, 7	"	"	"	"	"
J5	T4 "	26	6	23,24, 25,27, 28	3,4, 5,7, 8	"	"	"	"	"
J6	T6 "	27	7	24,25, 26,28, 29	4,5, 6,8, 9	"	"	"	"	"
J7	T7 "	28	8	26,27, 29,31, 32	6,7, 9,11, 12	"	"	"	"	"
J8	T9 "	29	9	27,28, 31,32, 33	7,8, 11,12, 13	"	"	"	"	"
J9	T11 "	31	11	28,29, 32,33, 34	8,9, 12,13, 14	"	"	"	"	"

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TABLE IX. Input crosstalk (high-level and low-level) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
J10	T8 Input cross-talk, high level	32	12	28,29, 31,33, 34	8,9, 11,13, 14	A	0	300	mV	13,14
J11	T10 "	33	13	29,31, 32,34, 35	9,11, 12,14, 15	"	"	"	"	"
J12	T12 "	34	14	31,32, 33,35, 36	11,12, 13,15, 16	"	"	"	"	"
J13	T13 "	35	15	33,34, 37,38, 39	13,14, 17,18, 19	"	"	"	"	"
J14	T15 "	36	16	34,35, 37,38, 39	14,15, 17,18, 19	"	"	"	"	"
J15	T17 "	37	17	35,36, 38,39, 40	15,16, 18,19, 20	"	"	"	"	"
J16	T14 "	38	18	35,36, 37,39, 40	15,16, 17,19, 20	"	"	"	"	"
J17	T16 "	39	19	35,36, 37,38, 40	15,16, 17,18, 20	"	"	"	"	"
J18	T18 "	40	20	35,36, 37,38, 39	15,16, 17,18, 19	"	"	"	"	"

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TABLE IX. Input crosstalk (high-level and low-level) test - Continued.

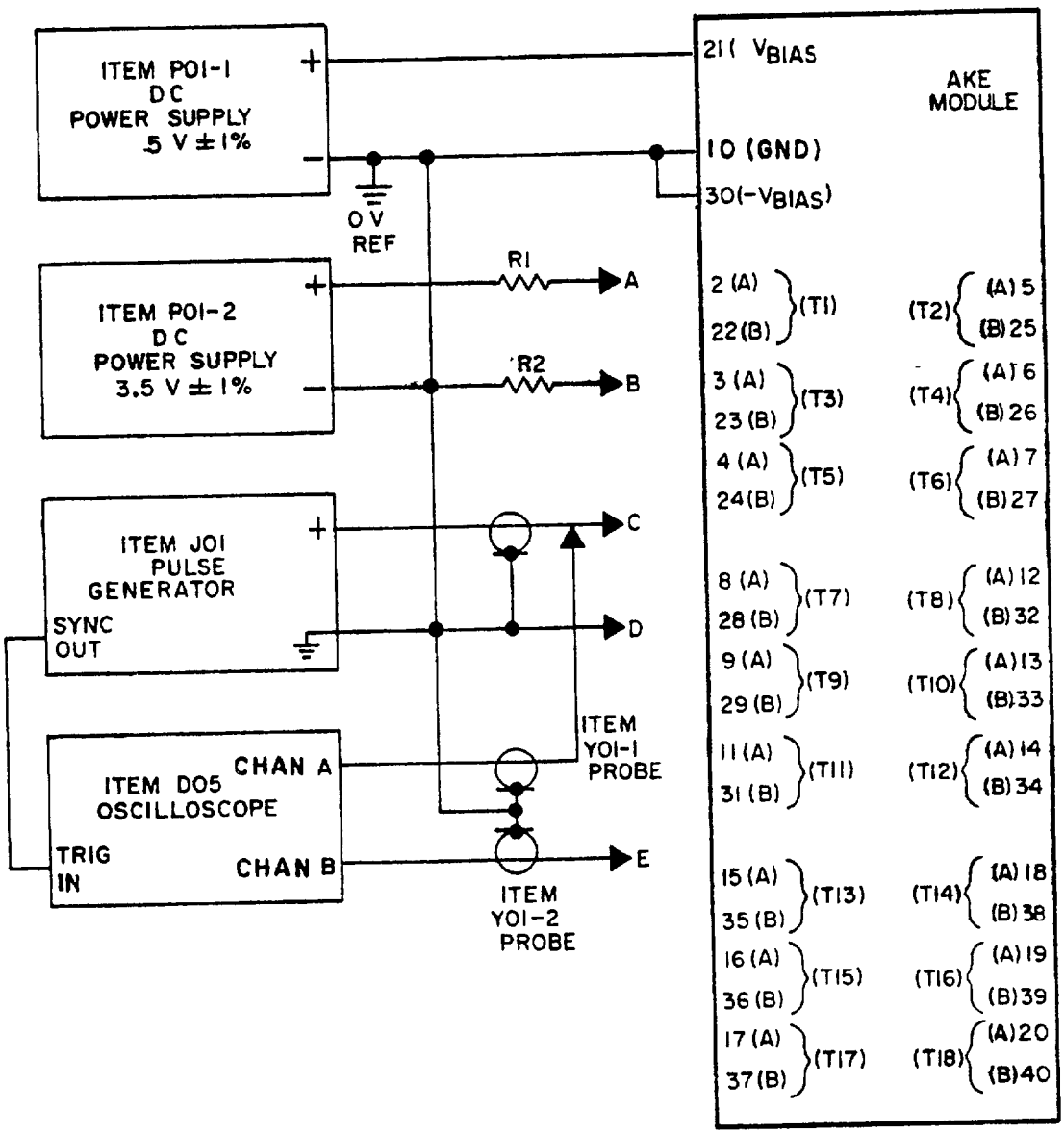
Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
K1	T1	Input cross-talk, low level	22	2	23,24, 25,26, 27	3,4, 5,6, 7	B	0	300	mV	13,14
K2	T3	"	23	3	22,24, 25,26, 27	2,4, 5,6, 7	"	"	"	"	"
K3	T5	"	24	4	22,23, 25,26, 27	2,3, 5,6, 7	"	"	"	"	"
K4	T?	"	25	5	22,23, 24,26, 27	2,3, 4,6, 7	"	"	"	"	"
K5	T4	"	26	6	23,24, 25,27, 28	3,4, 5,7, 8	"	"	"	"	"
K6	T6	"	27	7	24,25, 26,28, 29	4,5, 6,8, 9	"	"	"	"	"
K7	T7	"	28	8	26,27, 29,31, 32	6,7, 9,11, 12	"	"	"	"	"
K8	T9	"	29	9	27,28, 31,32, 33	7,8, 11,12, 13	"	"	"	"	"
K9	T11	"	31	11	28,29, 32,33, 34	8,9, 12,13, 14	"	"	"	"	"

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TABLE IX. Input crosstalk (high-level and low-level) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
K10	T8 Input cross-talk, low level	32	12	28,29, 31,33, 34	8,9, 11,13, 14	B	0	300	mV	13,14
K11	T10 "	33	13	29,31, 32,34, 35	9,11, 12,14, 15	"	"	"	"	"
K12	T12 "	34	14	31,32, 33,35, 36	11,12, 13,15, 16	"	"	"	"	"
K13	T13 "	35	15	33,34, 37,38, 39	13,14, 17,18, 19	"	"	"	"	"
K14	T15 "	36	16	34,35, 37,38, 39	14,15, 17,18, 19	"	"	"	"	"
K15	T17 "	37	17	35,36, 38,39, 40	15,16, 18,19, 20	"	"	"	"	"
K16	T14 "	38	18	35,36, 37,39, 40	15,16, 17,19, 20	"	"	"	"	"
K17	T16 "	39	19	35,36, 37,38, 40	15,16, 17,18, 20	"	"	"	"	"
K18	T18 "	40	20	35,36, 37,38, 39	15,16, 17,18, 19	"	"	"	"	"

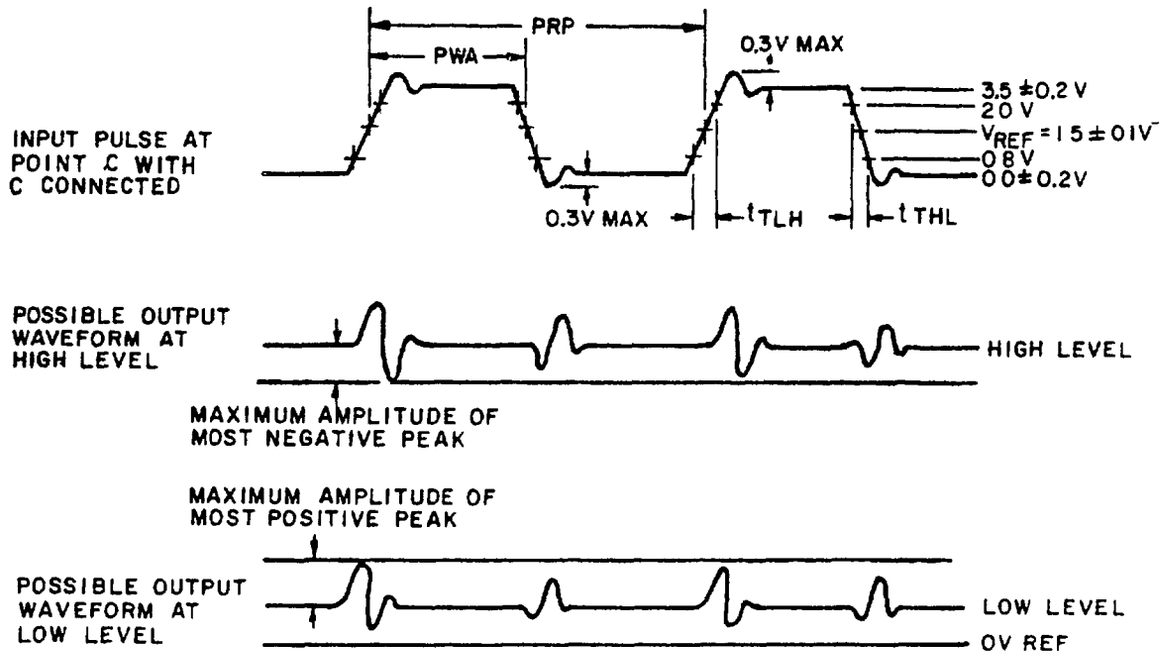
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NOTES:
1. R1 = 49.9Ω ±1%, 1/8 W.
2. R2 = 12.7Ω ±1%, 1/8 W.

* FIGURE 13. Input crosstalk (high-level and low-level) test circuit for table IX.

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NOTES:
1. Input pulse parameters:

   (a) PRP = 1,000 ±200 ns.
   (b) PWA = 150 ±50 ns.
   (c) $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns.

   For each test condition with the module connected to item J01, adjust item J01 for the required pulse amplitude, pulse low level voltage, and transition times. Verify the input pulse parameters at each pin before making each measurement.
2. Physical length of electrical conductors between item J01 cable and point C; items Y01-1 and Y01-2 to input pins; and ground leads to test fixture and module ground pins shall be less than 1 inch (25.4 mm).

* FIGURE 14. Input corsstalk (high-level and low-level) test cirucit waveforms for table IX.

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